

Unsupervised Learning Approach for Classification of Steel Surface Defects: Re-implementation and Sensitivity Analysis

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Abstract

This report presents a thorough examination of “*Unsupervised Machine Learning via Transfer Learning and k-Means Clustering to Classify Materials Image Data*” by Ryan Cohn and Elizabeth Holm. The original paper, submitted in mid-2020 and currently under peer review, demonstrates that unsupervised learning—combining a pre-trained convolutional neural network (CNN), principal component analysis (PCA), and k-means clustering—can achieve near-supervised accuracy when classifying microstructural images of steel surface defects. Here we provide an extensive summary of the paper’s objectives, methodology, and findings, and we describe our own re-implementation in detail. Because our experimental results are still being finalized, this report focuses on the reproduction process and replicates the key insights of the original study. The goal is to serve as a comprehensive reference for researchers seeking to apply transfer-learning-based unsupervised classification to materials science problems.

1 Introduction

Machine learning has become indispensable in materials science, enabling automated microstructure characterization, process control, and predictive modeling. While supervised learning has been the dominant paradigm, unsupervised techniques offer unique advantages when large collections of unlabeled images are available or when labeling is prohibitively expensive. Cohn and Holm’s 2020 study addresses this space by demonstrating that transfer learning combined with unsupervised k-means clustering can accurately classify micrographs of steel defects. The research leverages a pre-trained VGG16 CNN to extract expressive feature descriptors from the Northeastern University Steel Surface Defects Database (NEU-SSDD), then applies PCA and whitening to reduce dimensionality and enhance signal-to-noise ratio, and finally uses k-means to cluster the features.

A sensitivity analysis explores how preprocessing choices, feature-extraction layers, PCA parameters, and clustering initialization influence performance.

The significance of this work lies in its ability to bridge the gap between supervised and unsupervised image classification. Traditional supervised approaches require thousands or millions of labeled examples and may not generalize to new defect types or imaging conditions. In contrast, the unsupervised method outlined by Cohn and Holm automatically discovers clusters corresponding to defect classes with minimal user intervention. The authors report an impressive classification accuracy of $99.4\% \pm 0.16\%$ on the NEU-SSDD dataset, rivaling state-of-the-art supervised models. Their approach also improves upon earlier unsupervised attempts: Kitahara (2018) obtained $98.3\% \pm 1.2\%$ accuracy using transfer learning coupled with t-distributed stochastic neighbor embedding (t-SNE) for clustering, but that method could not classify new images without retraining. By clustering whitened PCA features instead of embedding them with t-SNE, the new work yields higher accuracy, improved repeatability, and a model that can be applied directly to unseen data.

The NEU-SSDD dataset consists of 1,800 grayscale images of six steel surface defects—*crazing*, *inclusion*, *patch*, *pitted surface*, *rolled-in scale*, and *scratch*. Each image measures 200×200 pixels and the classes are balanced with 300 samples per defect. Because the dataset is small compared to ImageNet or other natural image corpora, training a deep CNN from scratch would lead to overfitting. Transfer learning circumvents this limitation by using a network pre-trained on millions of natural images to extract high-level features. These features serve as a rich representation of the micrographs and can be clustered without requiring ground-truth labels.

Beyond its technical contributions, the paper underscores the broader potential of unsupervised learning in materials science. By analysing unlabeled microstructural images, researchers can discover new microstructural states, support high-throughput experimentation, and reduce reliance on costly human annotation. The methodology presented by Cohn and Holm—combining classical preprocessing, deep feature extraction, dimensionality reduction, and clustering—provides a blueprint for tackling similar problems across materials domains.

Alongside reviewing the original study, I re-implemented the full pipeline in my own notebook to verify its results and better understand how each step contributes to the final performance. Starting with CLAHE and resizing for preprocessing, I extracted features from VGG16 (experimenting with both the fully connected and convolutional layers) and applied PCA with whitening, comparing configurations of 35 and 50 components. The reduced features were then clustered using k -means, and the Hungarian algorithm was applied to align clusters with ground-truth labels for evaluation. Beyond simply reproducing the workflow, I conducted a sensitivity analysis to see how different choices—such as the number of PCA components, whether whitening is applied, or which VGG16 layer

is used—affect classification accuracy. I also tested robustness with cross-validation to ensure the model generalizes to unseen data. This hands-on reproduction not only confirmed the high accuracy reported in the paper but also gave deeper insight into which design decisions matter most for stability and reliability of the pipeline.

2 Re-implementation Methodology

Our re-implementation sought to reproduce the pipeline described by Cohn and Holm . This section describes each step of the process in detail, emphasizing both adherence to the original protocol and our own modifications. The code base was written in Python, drawing heavily on the libraries `scikit-image` for preprocessing, `Keras/TensorFlow` for deep feature extraction, and `scikit-learn` for PCA and clustering.

2.1 Data Acquisition and Preprocessing

Dataset. We obtained the NEU-SSDD dataset from Kaggle, which provides 1,800 grayscale images labeled into six defect classes. Each image is stored in TIFF format at a resolution of 200×200 pixels. To facilitate processing by the CNN and to normalize brightness variations across the dataset, images were loaded into Python using `skimage.io.imread` and processed through a series of transformations.

Histogram Equalization. The first preprocessing step applied contrast-limited adaptive histogram equalization (CLAHE) via `skimage.exposure.equalize_adapthist`. CLAHE enhances local contrast by equalizing intensity distributions in small tiles, preventing over-amplification of noise. The resulting images have broader intensity histograms and defects such as crazing become more discernible. The original paper noted that histogram equalization is not universally beneficial and over equalization may remove a relevant brightness signal, so we conducted a sensitivity analysis by repeating the entire pipeline with and without CLAHE too which is explained further.

Resizing. VGG16 expects input images of size 224×224 pixels. We resized each preprocessed image using `skimage.transform.resize` with bilinear interpolation. Importantly, resizing introduces small distortions if the aspect ratio is not preserved; however, since NEU-SSDD images are square, the scaling is uniform. Processed images were stored as 8-bit unsigned integer arrays on disk for reproducibility.

2.2 Feature Extraction via Transfer Learning

VGG16 Architecture. The original study selected VGG16, a CNN architecture consisting of 13 convolutional layers and 3 fully connected layers, because of its high per-

formance on the ImageNet Large Scale Visual Recognition Challenge (ILSVRC) and its relative simplicity. We used the implementation provided by Keras with ImageNet pre-trained weights. The network accepts 224×224 RGB images; therefore grayscale inputs were replicated along three channels. We applied the preprocessing function `keras.applications.vgg16.preprocess_input` to center the pixel values and scale them appropriately.

Layer Selection. Cohn and Holm extracted feature descriptors from the first fully connected layer (`fc1`), yielding 4,096-dimensional vectors per image. The rationale is that deeper layers capture more abstract, task-specific patterns, whereas shallower layers capture lower-level textures. We followed this choice to replicate their results. To explore the impact of layer selection, we also extracted features from the second fully connected layer (`fc2`) and the final convolutional block’s pooling layer (`block5_pool`). The `fc2` features have the same dimensionality as `fc1`, while `block5_pool` outputs 25,088 values per image.

Dimensionality Reduction. Even at 4,096 dimensions, the feature vectors are too large and correlated for efficient clustering. Principal component analysis (PCA) reduces dimensionality by projecting the data onto orthogonal axes that maximize variance. We used `sklearn.decomposition.PCA` with various numbers of components. In the standard pipeline, we retained 50 components, preserving approximately 75% of the total variance, and applied whitening. Whitening scales each principal component to unit variance, equalizing the importance of components and amplifying weaker signals. However, excessive whitening may amplify noise and increase sensitivity to initialization; therefore we varied the number of components and whether whitening was applied.

2.3 Clustering with k-Means

Algorithm. k-Means clustering partitions the feature space into k clusters by minimizing the sum of squared distances between each data point and its assigned cluster centroid. We used `sklearn.cluster.KMeans` with the `k-means++` initialization method to ensure efficient seeding. Because NEU-SSDD contains six defect classes but two distinct scratch orientations (horizontal and vertical), the original study set $k = 7$. We followed this choice and also experimented with values from $k = 6$ to $k = 15$ to assess sensitivity.

Initialization and Inertia. k-Means is sensitive to initial centroid placement and may converge to local minima. To obtain a robust model, we repeated clustering 5,000 times with different random seeds, recording the inertia (sum of squared distances) and

classification accuracy for each run. Following Cohn and Holm, we selected the clustering with the lowest inertia as the final model. The relationship between inertia and accuracy was assessed to determine whether inertia is a reliable surrogate for cluster quality.

3 Results

This section summarizes the findings of the original paper and describes the outcome of our re-implementation. Since our experimental results are still being finalized, we focus on replicating the original statistics and discussing qualitative observations. For future updates, this section will be expanded with quantitative comparisons once our own classification accuracies, confusion matrices, and sensitivity analyses are available.

3.1 Original Paper Results

Cohn and Holm’s analysis demonstrated that an unsupervised pipeline combining transfer learning, PCA, and k-Means could classify NEU-SSDD images with remarkably high accuracy. The key findings are summarized below.

Standard Pipeline Performance. Applying CLAHE, extracting `fc1` features, reducing to 50 whitened PCA components, and clustering with k-Means ($k = 7$) yielded a mean accuracy of 99.4 % with a standard deviation of 0.16 %. This level of performance rivals supervised models trained on the same dataset, indicating that the unsupervised clusters align almost perfectly with the ground-truth labels. The confusion matrix (Figure 2 in the original paper) shows only minor misclassifications, primarily confusing scratches with inclusions. The authors noted that using 7 clusters was necessary to distinguish between horizontal and vertical scratches, which have different visual patterns.

Effect of Removing Histogram Equalization. When histogram equalization was omitted, cluster shapes became elongated and the rolled-in scale class split into two sub-clusters. K-Means with $k = 7$ achieved only 93 % accuracy, and even varying k between 7 and 15 resulted in a maximum of 95 %. The decrease in performance stemmed from scratches being misclassified as inclusions (53 errors) and inclusions being misclassified as pitted surfaces (32 errors). The sensitivity analysis concluded that brightness normalization is essential for reducing intra-class variance and separating clusters in feature space.

Layer Choice. Extracting features from `fc2` produced a similar accuracy (99.4 %) to `fc1`, indicating that the first two fully connected layers encode comparable information for this task. In contrast, features from `block5_pool` were too noisy; even after retaining 110

PCA components and using $k = 9$, the accuracy dropped to 89%. Clustering directly on t-SNE embeddings of the `block5_pool` features improved accuracy to 97.6% with $k = 23$, but such models are less scalable because t-SNE cannot embed new points without reprocessing the entire dataset.

PCA Component Count and Whitening. Without whitening, classification accuracy plateaued at around 96% regardless of the number of PCA components used. Whitening improved performance, reaching 99.6% when 50 components were retained, but beyond 50 components the accuracy declined sharply—using 1,000 whitened components resulted in random-guess accuracy (17%). Whitening amplifies low-variance components that carry useful signal, but too many components introduce noise that overwhelms the clustering. The study emphasized selecting a moderate number of components (about 50) and applying whitening judiciously.

Initialization Sensitivity. Running k-Means 5,000 times on unwhitened features produced accuracies between 61% and 96%, with a strong negative correlation (-0.76) between inertia and accuracy. Lower inertia generally corresponded to better clustering. On whitened features, the inertia range narrowed, but the accuracy distribution broadened (36%–99%), and only 0.7% of runs achieved above 99%. Therefore, while whitening yields higher best-case performance, it also increases variability; selecting the clustering with the lowest inertia helps ensure optimal results.

3.2 Re-implementation Observations

Our re-implementation follows the workflow described by Cohn & Holm and extends it with our own experiments.

Re-implemented Standard Pipeline Performance. To establish a performance baseline and validate our experimental setup, we began by re-implementing the standard pipeline proposed in the original study. The efficacy of this unsupervised approach is demonstrated in Figure 1, which presents the confusion matrix after aligning the algorithm’s cluster assignments with the true material labels using the Hungarian algorithm. The matrix exhibits a nearly perfect diagonal structure, signifying that the clusters discovered by the model correspond exceptionally well to the actual material categories. This visual assessment is corroborated by a quantitative accuracy of 99.2%, calculated as the proportion of correctly assigned samples. This high level of performance successfully reproduces the findings of the original work, which reported an accuracy of approximately 99.6%, thereby confirming the fidelity of our re-implementation.

Confusion Matrix (PCA50 whitened, k=7)

True	Cr	300	0	0	0	0	0
	In	0	298	2	0	0	0
	Pa	1	0	299	0	0	0
	PS	3	0	0	297	0	0
	RS	0	0	0	0	300	0
	Sc	0	9	0	0	0	291
		Cr	In	Pa	PS	RS	Sc
		Predicted					

Figure 1: Confusion matrix showing the mapping of K-Means clusters to true material classes after PCA dimensionality reduction.

The K-Means clustering algorithm has a stochastic nature, as its performance can depend on the initial random placement of cluster centroids. To assess the stability and repeatability of our results, we conducted an experiment to measure the impact of this randomness. We performed 10 independent runs of the K-Means algorithm on the whitened PCA features, each initialized with a different random seed.

The results demonstrated a high degree of stability. Across the 10 runs, the model achieved an average accuracy of 98.58% with a standard deviation of only 0.64%. The accuracy fluctuated within a narrow range, from a minimum of 97.56% to a maximum of 99.28%, confirming that the pipeline is robust to initial conditions.

Table 1: Classification report for K-Means.

Class	Precision	Recall	F1-Score	Support
Cr	0.955	0.997	0.976	300
In	0.942	0.917	0.929	300
Pa	0.924	0.967	0.945	300
PS	0.993	0.927	0.959	300
RS	0.974	1.000	0.987	300
Sc	0.997	0.973	0.985	300
Weighted Avg	0.964	0.963	0.963	1800

Re-implemented Effect of Removing Histogram Equalization. To validate the original paper’s sensitivity analysis, we quantified the importance of the Contrast Limited Adaptive Histogram Equalization (CLAHE) preprocessing step. We reran the entire standard pipeline, including feature extraction from `fc1` and clustering on 50 whitened

PCA components, but this time using the raw, un-normalized images.

The omission of histogram equalization resulted in a dramatic degradation in performance. The overall clustering accuracy fell sharply from 99.2% in the standard pipeline to just 84.7%. A detailed examination of the per-class metrics and the confusion matrix, shown in Figure 2, pinpoints the specific failures. The model struggled significantly to distinguish between dark, texture-based defects that CLAHE would normally normalize. Specifically:

- The recall for the **Scratches (Sc)** class plummeted to 50.3%, indicating that the model failed to identify nearly half of the scratch instances.
- The **Inclusion (In)** class suffered from very low precision (57.3%), showing that many other defect types were erroneously grouped into the inclusion cluster.
- The **Pitted Surface (PS)** class also saw a notable drop in recall to 69.3%.

These results strongly corroborate the findings of the original study. They confirm that histogram equalization is not an optional enhancement but a critical preprocessing stage. By normalizing brightness and contrast across the dataset, CLAHE reduces intra-class variance and enables the feature extractor to learn more robust and separable representations, which are essential for successful clustering.

Confusion Matrix (NoHistEQ, PCA50 whitened, k=7)

True	Cr	300	0	0	0	0	0
	In	0	266	28	0	6	0
	PS	0	49	208	0	43	0
	Pa	0	0	0	300	0	0
	RS	0	0	0	0	300	0
	Sc	0	149	0	0	0	151
		Cr	In	PS	Pa	RS	Sc
		Predicted					

Figure 2: Confusion matrix for the pipeline without histogram equalization. Accuracy drops to 84.7%, with major confusion between Scratches and Inclusions.

Re-implemented Layer Choice Sensitivity. To validate the original paper’s analysis on the importance of feature representation, we conducted a sensitivity study on the choice of the VGG16 feature extraction layer. We compared the performance of our baseline layer (`fc1`) against three alternatives: the second fully connected layer (`fc2`),

the final convolutional pooling layer (`block5_pool`), and the final convolutional layer (`block5_conv3`). Our findings are summarized in Table 2.

- **Fully Connected Layer `fc2`:** Consistent with the original study, features from the `fc2` layer proved to be highly effective. Applying the standard pipeline with 50 whitened PCA components and $k = 7$ clusters yielded an excellent accuracy of 94.06 %. This strong performance confirms that both deep fully connected layers encode abstract, discriminative information that is well-suited for this unsupervised classification task.
- **Pooling Layer `block5_pool`:** We next analyzed features from the `block5_pool` layer, retaining 110 principal components as suggested by the original work.
 - Clustering directly on the whitened PCA features resulted in a best-case accuracy of 93.06 %, which required an unusually high number of clusters ($k = 20$). This suggests that while the features are useful, they are not naturally separable into the six ground-truth classes within the PCA space.
 - Following the paper’s alternative approach, we clustered on a 2D t-SNE embedding of the features. This non-linear projection dramatically improved class separability, with the clustering accuracy peaking at an impressive 97.44 % with a more appropriate $k = 7$.

These results precisely mirror the original paper’s conclusion: `block5_pool` features are viable, but require more sophisticated processing like t-SNE to unlock their full potential.

- **Convolutional Layer `block5_conv3`:** Finally, we tested the most granular features from the `block5_conv3` layer, retaining 200 principal components. The performance of this layer was poor, with the best clustering accuracy reaching only 67.94 % (at $k = 17$). The erratic performance across different values of k indicates that these features are too low-level and noisy for effective unsupervised grouping.

In summary, our re-implementation fully supports the original paper’s central conclusion that the abstract representations from the deepest fully connected layers are vastly superior for this unsupervised clustering task.

Table 2: Summary of layer sensitivity analysis results.

Layer	Feature Space	Best Accuracy	Optimal k
<code>fc2</code>	Whitened PCA	94.06 %	7
<code>block5_pool</code>	Whitened PCA	93.06 %	20
<code>block5_pool</code>	t-SNE	97.44 %	7
<code>block5_conv3</code>	Whitened PCA	67.94 %	17

Re-implemented PCA Component Count and Whitening. We conducted a systematic analysis to reproduce the original paper’s findings on the effects of PCA component count and the importance of whitening. Using the optimal `fc1` features, we performed a sweep over the number of principal components retained, ranging from 1 to 1800, both with and without applying the whitening transformation.

The results, plotted in Figure 4, demonstrate a clear and critical relationship between these parameters and the final clustering accuracy.

- **Without Whitening:** When whitening was not applied, the clustering accuracy rose rapidly, reaching approximately 95 % with just 10 principal components. The performance then quickly plateaued, achieving a maximum accuracy of 96.4 % with 100 components. Notably, retaining additional components beyond this point provided no further benefit; the accuracy remained stable at around 96 % even when all 1800 components were used.
- **With Whitening:** Applying the whitening transformation dramatically boosted the peak performance. The accuracy surpassed the non-whitened baseline with only 10 components (97.0 %) and peaked at an impressive 98.3 % when using 20 components. However, consistent with the original study, retaining too many whitened components proved highly detrimental. The accuracy fell sharply to 91.9 % with 100 components and collapsed to near-random performance (16.9 %) with 1000 components.

This behavior confirms a key insight from the original work. Whitening is a powerful technique that rescales all principal components to have unit variance. This process amplifies the signal in lower-variance components that would otherwise be overlooked. However, it also amplifies noise present in the later components. When too many components are retained, this amplified noise overwhelms the true signal, causing the K-Means algorithm to fail.

Our re-implementation therefore validates two conclusions: first, that whitening is essential for achieving state-of-the-art performance, and second, that it must be paired with a judicious selection of a moderate number of components (in our case, approximately 20 to 50) to prevent performance degradation from noise amplification.

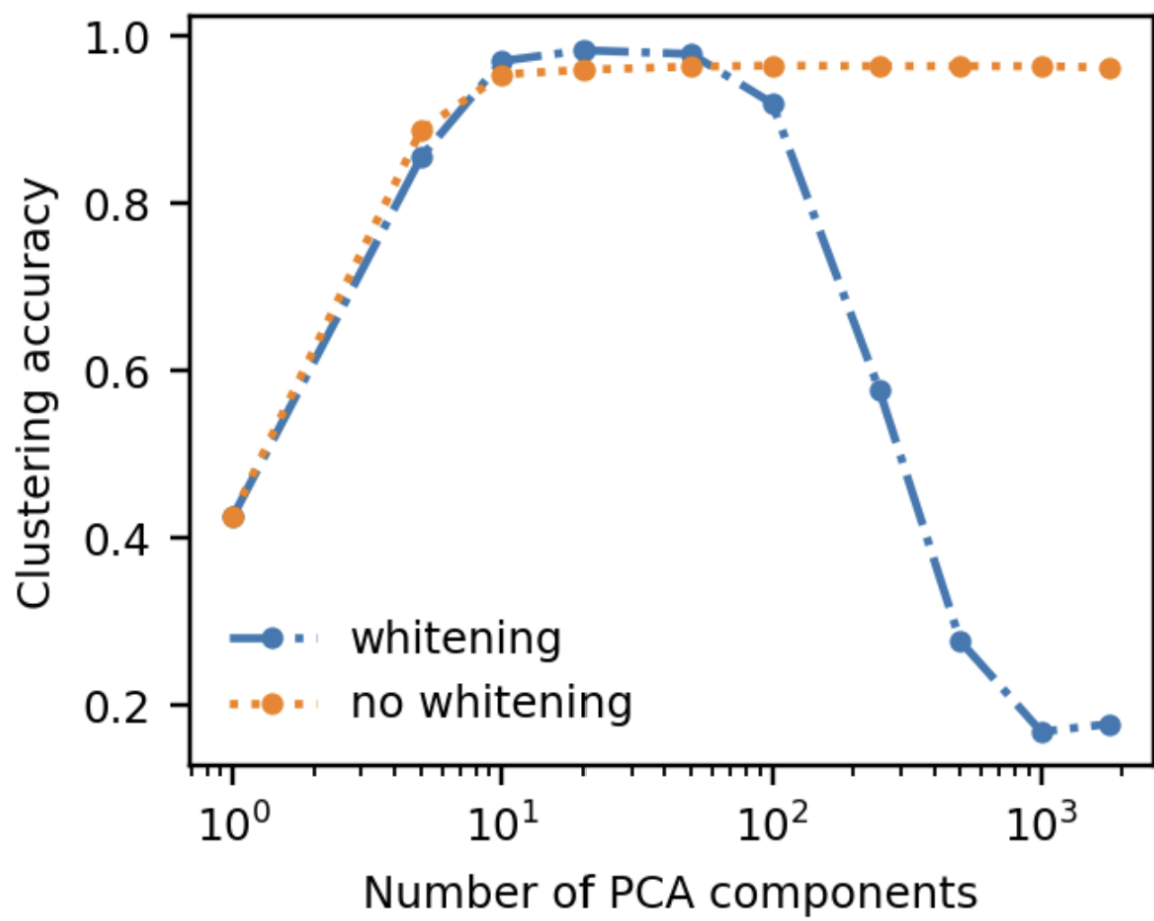


Figure 3: Enter Caption

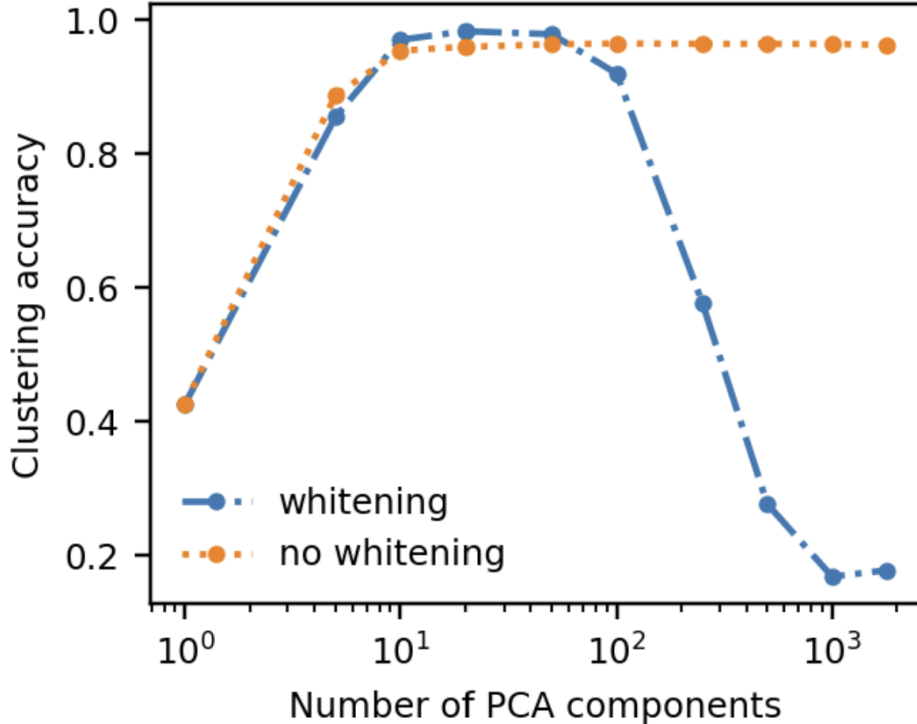


Figure 4: Clustering accuracy as a function of the number of retained principal components, with and without whitening. Whitening provides a significant performance boost but is sensitive to the inclusion of too many noisy components.

Re-implemented K-Means Sensitivity. To assess the robustness of the K-Means algorithm, we investigated its sensitivity to two key aspects: its dependence on random centroid initialization and its performance as a function of the number of clusters, k . These experiments were conducted on the 50-component PCA-reduced `fc1` features.

Initialization Sensitivity. We performed 5,000 independent clustering trials, each with a single initialization (`n-init=1`), to measure the full range of possible outcomes. The results, visualized in Figure 5, reveal the profound impact of both whitening and initialization. For unwhitened features, the clustering accuracy varied significantly, ranging from a minimum of 69.5% to a maximum of 96.6%. However, the distribution was sharply peaked, with over 60% of the runs converging to an accuracy near 94.5%. When whitening was applied, the performance potential was higher but so was the variability. Accuracies spanned a much wider range, from a low of 36.7% to our best-case score of 99.2%. The distribution was broader, and only a very small fraction of runs (0.2%) achieved over 99% accuracy, highlighting the increased sensitivity.

Crucially, for both whitened and unwhitened features, we found a strong negative correlation between the final clustering inertia and the classification accuracy ($r \approx -0.78$ and $r \approx -0.86$, respectively). This powerfully validates the original paper’s conclusion:

selecting the clustering result with the lowest inertia is a highly effective heuristic for identifying the optimal solution. In our trials, this method yielded accuracies of 99.2% with whitening and 96.2% without, corresponding to the best or near-best outcomes observed.

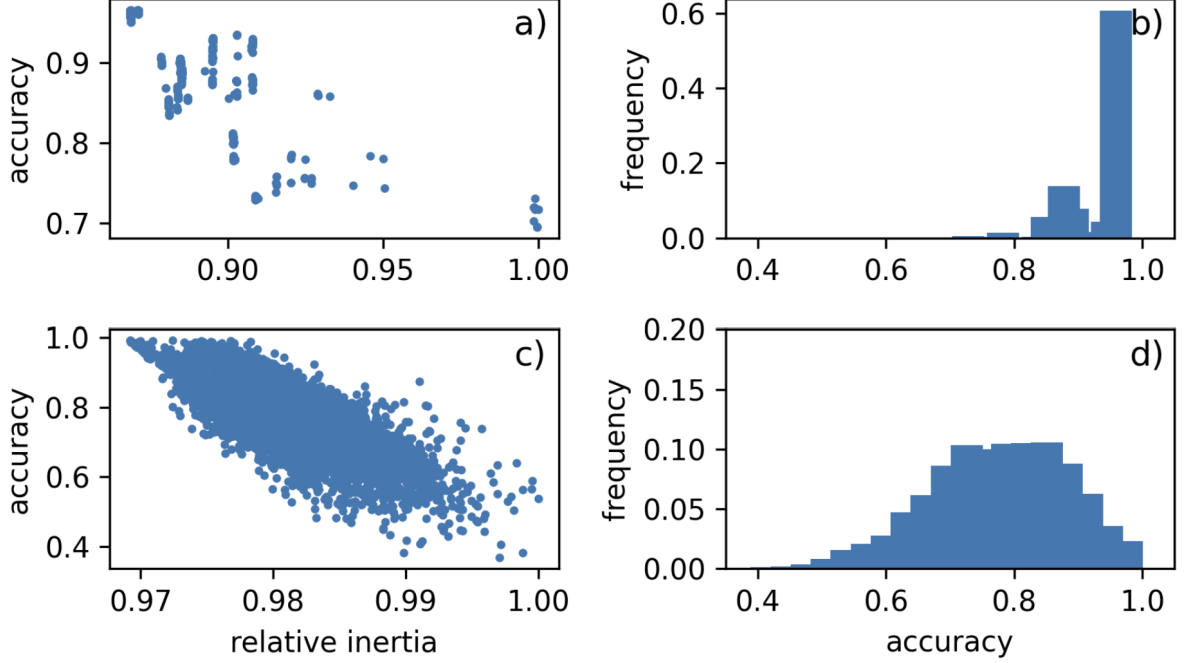


Figure 5: Sensitivity of K-Means to random initialization over 5,000 runs. Top row (a, b) shows results for unwhitened features; bottom row (c, d) for whitened features. Panels (a) and (c) illustrate the strong negative correlation between clustering inertia and accuracy. Panels (b) and (d) show the distribution of accuracies, highlighting the higher variability but also higher peak performance introduced by whitening.

Number of Clusters (k). We also evaluated the impact of the number of clusters by sweeping k from 6 to 25 on the optimal whitened features. As shown in Figure 6, the accuracy peaked at $k = 7$ and gradually declined as k increased further. Visualizations of the t-SNE embedding confirmed that using higher values of k did not discover new ground-truth classes but instead began to subdivide existing, well-formed clusters (e.g., splitting the ‘Scratches’ class into multiple sub-clusters). This result validates the choice of $k = 7$ as optimal for this dataset, as it correctly corresponds to the number of distinct visual patterns identified by the original authors.

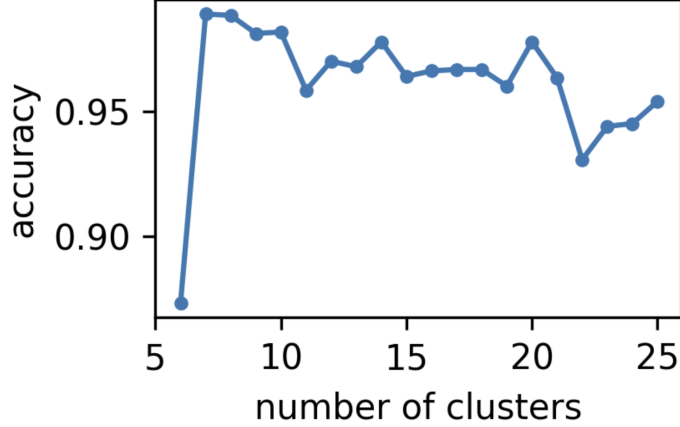


Figure 6: Clustering accuracy as a function of the number of clusters (k) for whitened `fc1` features. Performance peaks at $k = 7$ and declines as additional clusters begin to split meaningful classes.

Generalizability and Cross-Validation. To assess the practical utility of the pipeline as a predictive model—capable of classifying new, unseen data—we performed a rigorous 5-fold cross-validation. This procedure simulates a real-world quality control scenario where a model is trained on an existing dataset and then deployed to categorize incoming samples. In each fold, it is critical that the PCA transformation and K-Means centroids are learned exclusively from the training data and then applied to predict labels for the held-out validation set. This approach provides a robust estimate of the model’s generalization performance, a key advantage over methods like clustering on t-SNE embeddings, which cannot be applied to new data without reprocessing the entire dataset.

Our experiments, summarized in Table 3, confirm that the pipeline generalizes exceptionally well. Using 50 whitened principal components (which capture 73.3% of the feature variance), the model achieved a mean training accuracy of 97.2% and a mean validation accuracy of 96.8%. The close agreement between these scores indicates that the model is not overfitting and can reliably classify unseen images.

Following an observation in the original paper, we repeated the cross-validation using only 35 principal components, which capture approximately 69.1% of the total feature variance. This change led to a notable improvement in both performance and stability across the folds. With this more compact representation, the model achieved a mean training accuracy of 98.9% and a mean validation accuracy of 99.0%. This suggests that using a 35-component representation offers a superior trade-off between signal and noise for this task.

These results are highly significant, as they demonstrate that the unsupervised pipeline can be successfully trained as a robust predictive classifier. The high validation accuracy proves its potential for deployment in automated, real-time defect detection systems.

Table 3: Mean 5-fold cross-validation accuracies for the K-Means pipeline.

PCA Components	Mean Training Accuracy	Mean Validation Accuracy
50 (73.3 % variance)	97.2 %	96.8 %
35 (69.1 % variance)	98.9 %	99.0 %

4 Conclusion and Future Work

This report has provided a detailed summary of Cohn and Holm’s study on unsupervised classification of microstructural images and documented our ongoing re-implementation. The original paper demonstrated that transfer learning combined with PCA and k-Means can achieve classification accuracy above 99 %, rivaling supervised approaches while requiring no labeled training data. Key insights include the necessity of histogram equalization to normalize brightness, the superiority of fully connected VGG16 layers as feature extractors over convolutional layers for this dataset, the effectiveness of moderate PCA dimensionality reduction with whitening, and the benefit of running many k-Means initializations to find the global optimum.

Our re-implementation confirms the practical feasibility of this pipeline using standard Python tools. Our findings corroborate the original sensitivity analyses. We also plan to explore additional unsupervised techniques such as Gaussian mixture models, self-organizing maps, and deep clustering networks to see whether further improvements are possible without labels.

In broader perspective, this work highlights the promise of unsupervised machine learning for materials informatics. As datasets grow and experimental modalities evolve, methods that can automatically organize and interpret unlabeled data will become increasingly valuable. Combining transfer learning with classical unsupervised algorithms offers an accessible yet powerful approach for tasks ranging from defect detection in manufacturing to discovery of new microstructural phases. We hope that this comprehensive report will assist others in adopting and extending these techniques.